

Exercise series 1:

Entity Relationship Model

Introduction to databases



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Scenario

- *Construct a model for a car insurance company whose customers own one or more cars each.*
- *Each car has associated with it zero to any number of recorded accidents.*
- *Each insurance policy covers one or more cars, and has one or more premium payments associated with it.*
- *Each payment is for a particular period of time, and has associated due date, and the date the payment was received.*

Steps

1. Determine **entities**

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2. Determine **primitive attributes**

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2. Determine **primitive attributes**
3. Determine **attributes** that define the **identity** of an entity



Steps

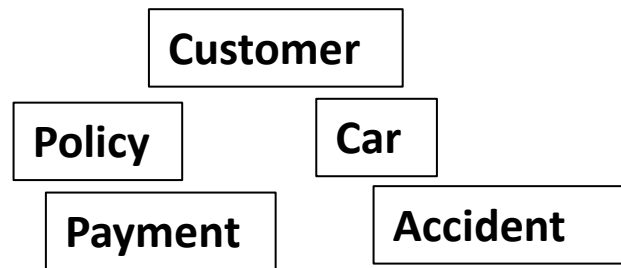
1. Determine **entities**
2. Determine **primitive attributes**
3. Determine **attributes** that define the **identity** of an entity
4. Determine **relations**

Steps

1. Determine **entities**
2. Determine **primitive attributes**
3. Determine **attributes** that define the **identity** of an entity
4. Determine **relations**
5. Determine **relation properties**: cardinality constraints (one/many) and if participation is partial or total (zero/one)

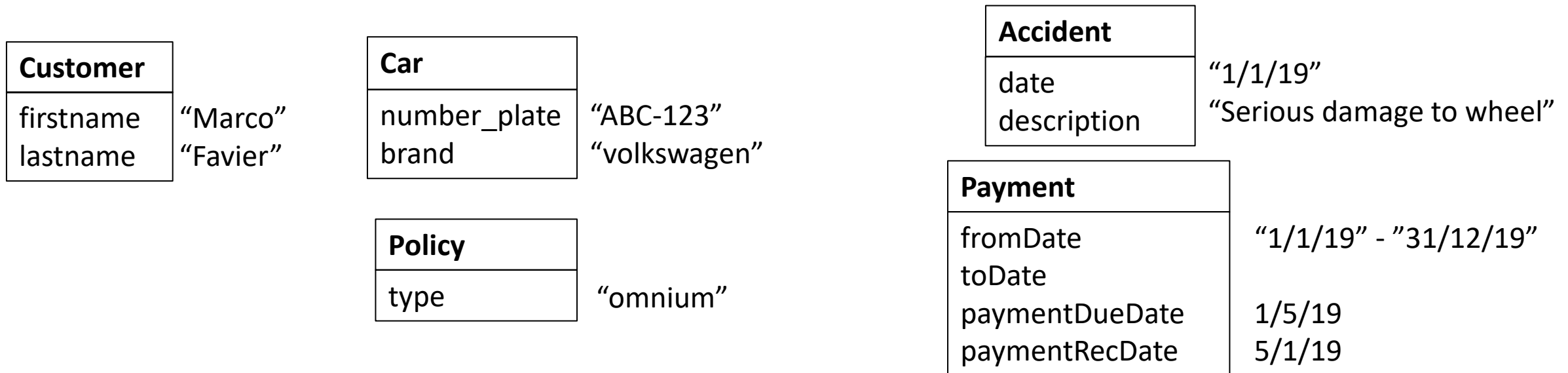
1> Determine **entities**

*Construct a model for a **car insurance company** whose **customers** own one or more **cars** each. Each car has associated with it zero to any number of recorded **accidents**. Each **insurance policy** covers one or more cars, and has one or more **premium payments** associated with it. Each payment is for a particular period of time, and has associated due date, and the date the payment was received.*



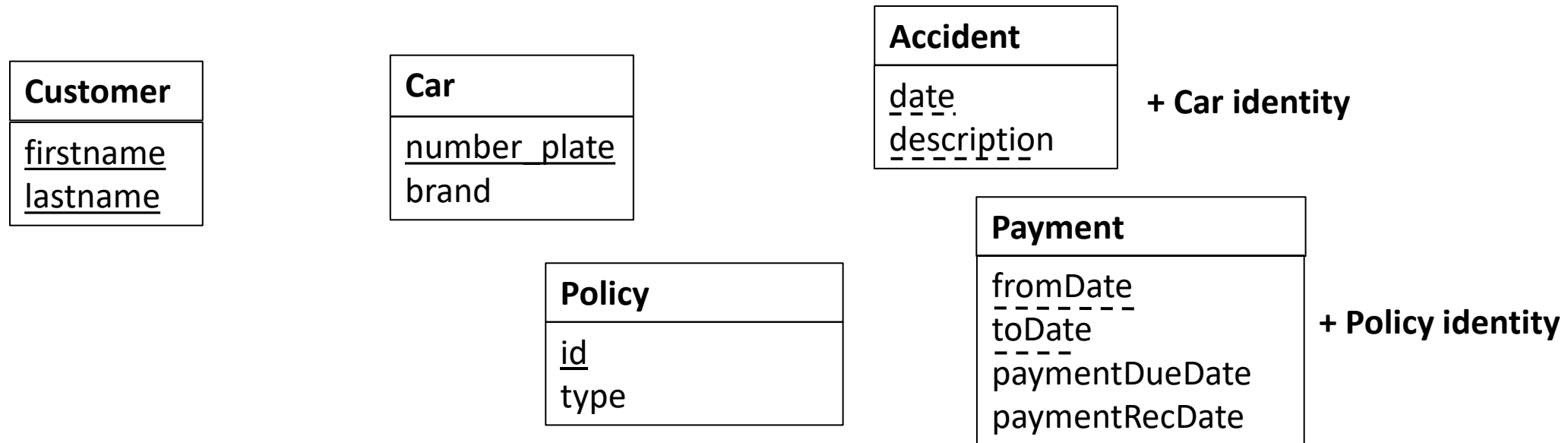
2> Determine primitive attributes

Construct a model for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars, and has one or more premium payments associated with it. Each payment is for a particular *period of time*, and has associated *due date*, and the *date the payment* was received.



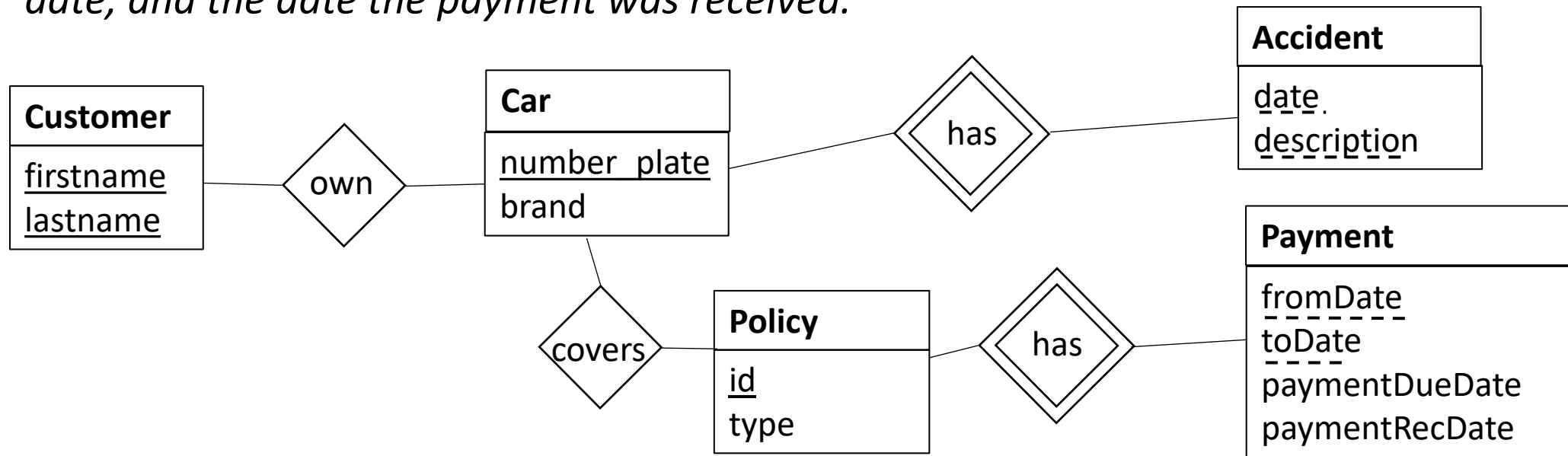
3> Determine **attributes** that define the **identity** of an entity

Construct a model for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars, and has one or more premium payments associated with it. Each payment is for a particular period of time, and has associated due date, and the date the payment was received.



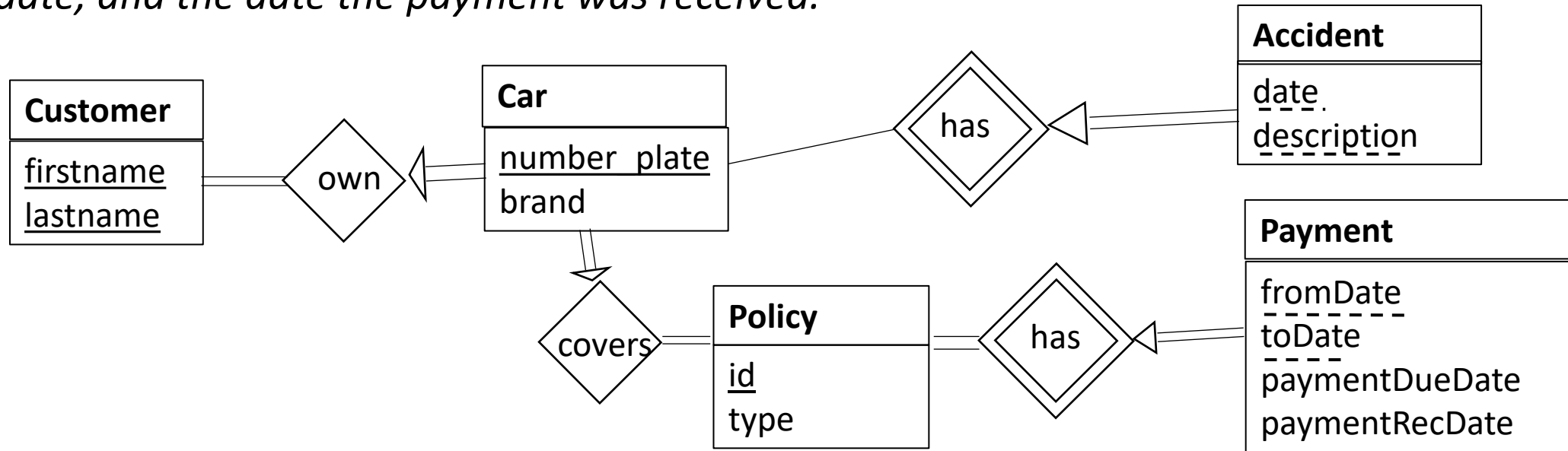
4> Determine relations

Construct a model for a car insurance company whose **customers own** one or more **cars** each. Each **car** has **associated** with it zero to any number of recorded **accidents**. Each insurance **policy covers** one or more **cars**, and has one or more **premium payments associated** with it. Each payment is for a particular period of time, and has associated due date, and the date the payment was received.



5> Determine **relation properties**: cardinality constraints (one/many) and if participation is partial or total (zero/one)

*Construct a model for a car insurance company whose customers own **one or more** cars each. Each car has associated with it **zero to any** number of recorded accidents. Each insurance policy covers **one or more** cars and has **one or more** premium payments associated with it. Each payment is for a particular period of time, and has associated due date, and the date the payment was received.*



Optimize your model by asking the following questions:

- Can this concept be best represented by an entity, a relation or an attribute?
- Can we replace an n-ary relation with multiple binary relations?
- Is the identity of this (weak) entity dependent on another entity?
- Is the same information represented in two ways?